Module-3

3.11 Assessment Questions

Part A - Multiple Choice Questions (1 Mark Each)

Q1. Which task is an example of text generation?

a. Clustering customers

b. Writing a summary

c. Sorting numbers

d. Encrypting a file

Answer: (b)

Q2. Which prompt element is especially important for image generation?

a. Lighting and composition

b. CPU speed

c. IP address

d. Database port

Answer: (a)

Q3. A model that accepts both an image and a text question is called:

a. Unimodal

b. Multimodal

c. Relational

d. Deterministic

Answer: (b)

Q4. Synthetic data is mainly used to:

a. Delete all real data

b. Create artificial records for testing/training

c. Replace all model evaluation

d. Remove the need for validation

Answer: (b)

Q5. Which is an advanced prompting technique?

a. Few-shot prompting

b. Hard disk formatting

c. Cable routing

d. Screen mirroring

Answer: (a)

Q6. Responsible AI includes which principle?

a. Fairness

b. Overfitting only

c. Randomness only

d. File compression

Answer: (a)

Q7. A guardrail placed before a model call is called:

a. Input guardrail

b. Output guardrail

c. Rendering guardrail

d. Database guardrail

Answer: (a)

Q8. Which risk means an LLM produces fluent but unsupported information?

a. Compression

b. Hallucination

c. Normalization

d. Tokenization

Answer: (b)

Part B – Short Answer Questions (5 Marks Each)

Q9. Explain text generation with a neat workflow diagram and two real-world applications.

Text generation is the process of using a generative AI model to create human-like text from a given prompt or input.

Workflow:

User Prompt

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Text Generation Model

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Understand Input & Context

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Generate Text

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Check / Refine Output

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Final Text

Applications:

1. **Chatbots:** Generate answers to customer or student questions.
2. **Content creation:** Generate emails, articles, summaries, and product descriptions.

Q10. Write the main prompt elements required for effective image generation.

The main elements of an effective image-generation prompt are:

1. **Subject:** What should be shown?
   Example: A girl reading a book.
2. **Environment/Background:** Where is the subject?
   Example: In a library.
3. **Style:** Type of visual style required.
   Example: Realistic, cartoon, anime.
4. **Composition/Camera:** Position and view of the image.
   Example: Close-up, wide-angle, aerial view.
5. **Lighting and Colors:** Specify light and color requirements.
   Example: Warm lighting and bright colors.
6. **Quality/Details:** Specify desired quality.
   Example: High-resolution, detailed image.

Q11. Differentiate between unimodal and multimodal generative models with examples.

Unimodal	Multimodal
Works mainly with one type of data.	Works with multiple types of data.
Input and output are usually from the same modality.	Can understand or generate text, images, audio, etc.
Less flexible.	More flexible and capable.
Example: A text-only language model.	Example: GPT-4o / GPT-5.6 Luna.
Used for tasks like text generation.	Used for tasks like image understanding and text generation together.

Example:
A text model generates an article from a text prompt, while a multimodal model can analyze an image and explain what is present in it.

Q12. What is synthetic data? Explain any four benefits and four risks.

Synthetic data is artificially generated data that resembles real-world data and is created using computers or AI models.

Benefits:

1. **Privacy:** Reduces the need to use sensitive real data.
2. **Large datasets:** Can generate large amounts of training data.
3. **Cost-effective:** Cheaper than collecting some real-world data.
4. **Rare cases:** Can create examples of uncommon or difficult situations.

Risks:

1. **Bias:** Synthetic data may contain the biases of the original data or model.
2. **Inaccuracy:** Generated data may not represent reality correctly.
3. **Privacy leakage:** Poorly generated synthetic data may still reveal information about real people.
4. **Poor quality:** Errors in synthetic data can lead to incorrect AI models.

Q13. Explain few-shot prompting, system instructions, and structured output prompting.

1. Few-shot prompting:
Giving the AI a few examples before asking it to perform a task.

Example:

Positive → "I love this product."

Negative → "This product is terrible."

Review → "The product is useful."

Answer → ?

2. System instructions:
Instructions that define how the AI should behave throughout the conversation.

Example:
“Answer all questions in simple language and provide short explanations.”

3. Structured output prompting:
Asking the AI to return information in a fixed format such as JSON, a table, or specific fields.

Example:

Name: Rahul

Age: 20

Course: BCA

These methods make AI responses more **consistent, controlled, and useful**.

Q14. What are Responsible AI principles? Explain fairness, privacy, transparency, and human oversight.

Responsible AI means developing and using AI in a safe, ethical, and responsible manner.

1. **Fairness:** AI should not unfairly discriminate against people based on characteristics such as gender, race, or background.
2. **Privacy:** Personal and sensitive information should be protected and used only when necessary and permitted.
3. **Transparency:** Users should understand how and why AI produces important decisions or outputs.
4. **Human Oversight:** Humans should monitor and review AI, especially when AI decisions can significantly affect people.

Goal: Responsible AI helps make AI systems **safe, trustworthy, fair, and accountable**.

Q15. Define generative AI guardrails. Explain input, output, retrieval, and human guardrails.

Generative AI guardrails are rules and controls used to prevent an AI system from producing harmful, unsafe, incorrect, or inappropriate results.

1. **Input Guardrails:** Check user input before sending it to the AI.
   Example: Detect harmful or inappropriate requests.
2. **Output Guardrails:** Check the AI's response before showing it to the user.
   Example: Block offensive or unsafe content.
3. **Retrieval Guardrails:** Control which documents or information the AI can access.
   Example: Allow a college chatbot to retrieve only official college documents.
4. **Human Guardrails:** A human reviews important or risky AI outputs.
   Example: A doctor reviews an AI-generated medical recommendation.

Q16. Why is human review important in high-stakes generative AI applications?

Human review is important because AI can produce **wrong, biased, or unsafe information**.

In high-stakes areas such as **healthcare, banking, law, and education**, incorrect AI decisions can seriously affect people.

Human review helps to:

1. **Detect errors** in AI-generated answers.
2. **Identify bias** and unfair decisions.
3. **Protect users** from harmful recommendations.
4. **Provide accountability** for important decisions.
5. **Make the final decision** when human judgment is necessary.

Long-Answer:-

Q17. Applications of Generative AI

Generative AI is AI that can create new content such as text, images, and data.

1. Text Generation

It creates human-like text.

- Used in chatbots
- Email writing
- Summaries
- Content creation

Example: College chatbot answering student questions.

2. Image Generation

It creates images from text prompts.

- Used in advertising
- Designing
- Games
- Posters

Example: Creating a college poster from a text description.

3. Multimodal Understanding

It can understand different types of data like **text, images, and audio**.

Example: Uploading a diagram and asking AI to explain it.

4. Synthetic Data

It creates artificial data similar to real data.

Example: Creating fake bank transactions to test a fraud detection system.

Conclusion: Generative AI is useful for **communication, design, analysis, and data generation**.

Q18. Advanced Prompt Engineering

Prompt engineering means creating good instructions to get better answers from AI.

Main techniques:

1. **System Instructions:** Tell the AI how to behave.
   Example: "Answer in simple English."
2. **Prompt Templates:** Use a fixed format for repeated tasks.
   Example: "Summarize this topic in {number} words."
3. **Few-Shot Examples:** Give examples to show what answer is expected.
4. **Structured Output:** Ask for output in a fixed format such as a table or JSON.
5. **Prompt Chaining:** Break one big task into smaller steps.
   Example: Read → Summarize → Check → Answer.
6. **Grounding:** Give trusted documents or information to the AI so it does not guess.
7. **Evaluation:** Check the answer for **accuracy, relevance, and safety**.

Conclusion: Good prompt engineering makes AI responses more **accurate and useful**.

Q19. Google Cloud College FAQ Chatbot

A college can create an AI chatbot to answer student questions.

Architecture

Student

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Chatbot

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Search College Documents

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AI Model

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Answer

1. Data Sources

The system uses:

- College circulars
- Exam notices
- Academic rules
- Course information
- FAQs

2. Prompt

The prompt can say:

“Answer only using official college documents. If the answer is not found, say ‘Information not found.’”

3. Model

A **Google Cloud generative AI model** generates the final answer.

4. Safety

- Check user input
- Protect student information
- Filter unsafe content
- Do not allow guessing

5. Monitoring

Check:

- Number of questions
- Response time
- Errors
- User feedback

6. Evaluation

Test the chatbot for:

- Accuracy
- Relevance
- Safety
- User satisfaction

Conclusion: The chatbot gives students **quick and reliable answers from official college information**.

Q20. Responsible AI Challenges

Generative AI has some important risks.

Challenge	Simple Meaning	Solution
Hallucination	AI gives false information	Fact-check and use trusted sources
Bias	AI gives unfair answers	Use fair and diverse data
Privacy	Personal information may leak	Protect sensitive data
Unsafe Content	AI may generate harmful content	Use safety filters
Copyright	AI may create content similar to protected work	Use legal/licensed data
Overreliance	People may blindly trust AI	Human checking

Conclusion: AI should be used with **safety rules, monitoring, and human supervision**.

Q21. Layered Guardrails

Guardrails are safety controls that prevent an AI system from giving harmful or incorrect results.

Workflow

User Input

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Input Check

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AI Model

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Output Check

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Human Review

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Final Answer

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Monitoring

1. Pre-processing

Check the input before sending it to AI.

Example: Block harmful requests.

2. Model Configuration

Give the AI proper instructions and restrict access to sensitive information.

3. Post-processing

Check the AI's answer before showing it.

Example: Remove harmful or private information.

4. Human Review

A human checks important or high-risk answers.

Example: A doctor checks an AI medical answer.

5. Monitoring

Continuously check:

- Errors

- Safety problems
- User feedback
- Answer quality

Case-study:-

Q22. Case Study – University AI Assistant (5 Marks)

1. Suitable Generative AI Capabilities

The system should have:

- **Text generation** – answer student questions.
- **Question answering** – understand student queries.
- **Document retrieval** – find information from approved documents.
- **Summarization** – explain long rules in simple language.
- **Multilingual support** – answer in different languages if required.

2. Prompt Template and Grounding

Prompt template:

“You are a university assistant. Answer only using the provided approved documents. Do not make up information. If the answer is not found, say ‘Information not available in approved documents.’ Never reveal personal data.”

Grounding:
Store approved university documents and retrieve the relevant document before generating an answer.

3. Input and Output Guardrails

Input guardrails:

- Block harmful or inappropriate questions.
- Detect personal/sensitive information.
- Check user access.

Output guardrails:

- Check for personal data.
- Block harmful information.
- Check that the answer is supported by documents.
- Prevent hallucinations.

4. Responsible AI Principles

- **Fairness:** Give the same information to all students.
- **Privacy:** Do not reveal student personal data.
- **Transparency:** Tell users when information is unavailable.
- **Human Oversight:** Important issues should be reviewed by university staff.

5. Evaluation Metrics

Check:

- **Accuracy**
- **Relevance**
- **Groundedness**
- **Privacy protection**
- **Safety**
- **Response time**
- **User satisfaction**

Conclusion: The assistant should provide **accurate, document-based, safe, and privacy-protected answers**.

Q23. Compare and Analyse Generative AI Types

Dimension	**Text Generation**	**Image Generation**	**Multimodal**	**Synthetic Data**
Input type	Text	Text/image	Text, image, audio	Data/instructions
Output type	Text	Image	Text, image, audio	Artificial data
Common model	LLM	Diffusion model	Multimodal LLM	GAN / diffusion
Use cases	Chatbots, emails	Posters, designs	Image analysis, assistants	AI training, testing
Prompt requirements	Clear text instructions	Subject, style, lighting	Text + other media	Data type and rules

Dimension	Text Generation	Image Generation	Multimodal	Synthetic Data
Evaluation	Accuracy, relevance	Quality, similarity	Accuracy, grounding	Quality, privacy
Main risk	Hallucination	Copyright, harmful images	Privacy, hallucination	Bias, privacy
Guardrails	Content filters	Image safety filters	Input/output filters	Privacy and quality checks

Project Combining All Four

Example: Smart University Assistant

- **Text generation:** Answers student questions.
- **Image generation:** Creates posters for university events.
- **Multimodal AI:** Reads and explains notices/images.
- **Synthetic data:** Creates artificial student data for testing.